

NAVNEET SHANKAR

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PROFESSIONAL SUMMARY

Software Engineer with 3 years of experience skilled in C++, Go, Java, Python, Scala, JavaScript, TypeScript. Experienced with CUDA, LLVM IR, GPU Programming, scalable/distributed data systems (Hadoop, Spark, Kafka, Kubernetes, Docker, OpenShift, Airflow, AWS,GCP), and AI/ML methodologies (Transformers, GPT, PyTorch, Reinforcement Learning, Diffusion models, robotics simulation, auto-regressive models, world models).

EDUCATION

Virginia Tech

Master's, Computer Engineering

- Relevant Coursework: Deep Learning, Advanced Computer Vision, Multiprocessor Programming, Compiler Optimization, Network Architectures and Protocols.

Aug 2024 - Dec 2025

GPA: 3.6

National Institute of Technology Karnataka

Bachelor's, Electronics and Communication Engineering

Jul 2017 - May 2021

PROFESSIONAL EXPERIENCE

Virginia Tech

Research Assistant

- Working as a Research Assistant in Prof. Eli Tilevich's Lab, focusing on creating a Domain Specific Language for interfacing between applications on Distributed Systems and Replicating Data Libraries.

Blacksburg, VA, USA

Aug 2025 - Present

Citibank

Software Engineer - Olympus (Big Data Division)

- Optimized ETL processes amounting to around 30% in time savings through refined Spark data, pipelines which included writing time and memory efficient algorithms in scala and python which leveraged the parallelism of distributed systems/storage, significantly improving data ingestion speed and system reliability.
- Enhanced Hadoop HDFS distributed storage efficiency by 40%, reducing storage costs through analysis and performance tuning primarily by eliminating the small file problem.
- Built highly scalable Kafka streaming services processing over 3 TB data daily, ensuring seamless real-time data analytics.

Chennai, TN, India

Sep 2022 - Jul 2024

Citibank

Software Engineer - Commercial Credit Cards

- Developed RESTful microservices using Spring Boot mainly handling settlements, achieving 80% code coverage with Selenium tests.
- Led the development of a Eureka - based service discovery portal integrating dynamic scaling, load balancing and dynamic configuration for microservices. Involved building an API layer with Spring Boot and a React based UI.
- Led the development of a Resource management platform that was used for managing and allocating personnel and other resources for projects in the very large commercial cards team. Involved building a user interface with React, using a number of Spring boot microservices and accessing a PostgreSQL database as well as an Oracle SQL server database.

Chennai, TN, India

Jul 2021 - Sep 2022

SKILLS

Languages: C++, Java, Python, Scala, JavaScript, TypeScript, Kotlin, Rust, CUDA, Go, Typescript, React.js, Kotlin, C#, PHP, Node.js

Frameworks: Spring Boot, Selenium, Eureka, Apache Spark

AI/ML: Transformers, GPT, Pytorch, Tensorflow, LLM, Reinforcement Learning, Natural Language Processing (NLP), Diffusion Models, Autoregressive models

Big Data: Hadoop, Hadoop, Kafka, Spark, Apache Hive, Impala, Postgres, Airflow, MongoDB, Oracle SQL

Tools /Platforms: Kubernetes, Docker, OpenShift, Git, Linux/Unix, Jenkins, Bash scripting, JIRA, Bitbucket, CI/CD, AWS, Vim, EMACS, GCP, SaaS, OAuth

Networking & Security: TCP/IP, UDP, Wifi, Zigbee, Vulnerability Testing, CAN Bus, TLS

PROJECTS & OUTSIDE EXPERIENCE

4D Temporal Modeling using VQ-GAN for MRI

- Implemented a generative Vector Quantized GAN model in PyTorch for reconstructing temporal MRI data.
- Utilised CUDA mixed-precision training techniques, achieving 15% accuracy improvement and reduced memory usage by 30%.
- Vector quantization resulted in visible improvement in terms of sharpness and clarity of generated images and the SSIM metric as well.
- [Link to project](#)

Apr 2025 - May 2025

LLVM GPU Loop Tiling Optimization

- Designed LLVM IR compiler optimization passes for automated loop tiling in CUDA kernels , achieving a 3x performance speedup for GPUs.
- Created comprehensive benchmarking tools in python to facilitate efficient parameter tuning and performance analysis
- Potential use case for ML/AI training applications to improve memory and time complexity.
- [Link to project](#)

Mar 2025 - May 2025

RKHS Transformer for Sequence Modeling

- Integrated Reproducing Kernel Hilbert Space (RKHS) and Vector Quantization attention mechanisms within the GPT 2 Transformer architecture to improve NLP sequence modeling.
- Achieved significant improvements in perplexity (29.62) and near-zero embedding errors (MSE).
- [Link to project](#)

Apr 2025 - May 2025

Scene Prediction for Vehicular Paths

- Built advanced generative deep learning models (UNet based Autoencoders, GRUs, Attention mechanisms) to accurately predict future vehicular movement from video data.
- Successfully validated models with real-world dataset such as the KITTI dataset, demonstrating strong predictive accuracy and robustness.
- [Link to project](#)

Nov 2024 - Nov 2024

Multiprocessor ML Optimization

- Implemented efficient parallel algorithms using multithreading to accelerate PCA, KNN, Random Forest, and Neural Network computations.
- Developed benchmarking suite to rigorously evaluate performance across various models and dataset sizes.
- [Link to project](#)

Nov 2024 - Nov 2024